# Predicting FEMA Hazard Mitigation Costs and State Trends

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**Abstract.** This capstone project analyzes the FEMA Hazard Mitigation Assistance (HMA) Mitigated Properties dataset (v4) to uncover patterns in disaster mitigation spending and predict future project costs. Focusing on U.S. states post-2000, the study employs exploratory data analysis (EDA) to reveal skewed cost distributions, reactive funding trends, and geographic hotspots. Supervised regression models-Linear Regression (LR) and Random Forest Regressor (RFR)-predict log-transformed costs, with RFR selected for superior performance (test $R^2 = 0.5847$ on log scale). Projections for FY2030 indicate a 17.9% increase in national spending ($1.63B to $1.92B), but sharp declines in historical hotspots, signaling the risks of risk redistribution. Insights emphasize proactive, equitable funding to enhance resilience amid rising disasters.

**Keywords:** FEMA· Hazard Mitigation· Project Cost Analysis· Exploratory Data Analysis (EDA)· Linear Regression Forecasting· Random Forest Forecasting

# 1 Introduction:

## 1.1 Background

Disaster mitigation is an essential element of national resilience planning, particularly as the frequency and economic impact of natural disasters continue to increase in the United States [1]. The Federal Emergency Management Agency (FEMA) plays a key role in managing hazard mitigation programs aimed at reducing future disaster losses. Through its Hazard Mitigation Assistance (HMA) programs, FEMA funds projects that improve community infrastructure, prevent repeated damage, and lower long-term recovery costs through measures such as property acquisition, flood control, and infrastructure hardening [2] and [3].

Studies have shown that hazard mitigation efforts significantly reduce disaster related losses. For example, [4] demonstrated that every dollar invested in FEMA's mitigation grants yields multiple dollars in avoided future disaster costs. Their benefit-cost analysis found that FEMA-funded projects are cost-effective and socially beneficial regardless of the type of danger.

## 1.2 Significance

The project analyzes how FEMA Hazard Mitigation Assistance (HMA) project costs vary across US states post 2000, hazard types, and property characteristics. Identifying hotspots and predictors enables more targeted allocation of limited federal funds, enhancing preparedness in vulnerable regions. This problem is also technically valuable as there are not many studies that combine exploratory analysis and predictive modeling with visualization.

### 1.3 Goals of this Project

The primary objectives are:

- Conduct an EDA on post-2000 HMA data, identifying cost trends by fiscal year, state, and project characteristics.
- Build and evaluate regression models (Linear Regression and Random Forest) to predict project-level mitigation costs.
- Visualize patterns and recommend strategies for equitable funding.

## 2 Data Description:

### 2.1 Data Source, Format, and Size

The FEMA Hazard Mitigation Assistance (HMA) dataset was obtained from the FEMA Open Data Portal. It is provided in CSV format, approximately 15 MB in size, containing 97,538 records and 19 columns in its original form. For this study, the dataset was filtered to include only U.S. states and projects completed after fiscal year 2000, resulting in 84,026 records across 11 key variables.

Data preparation steps standardized categorical variables, handled missing fields, and derived new attributes such as *totalMitigationCost*. These transformations ensured a consistent, analysis-ready dataset suitable for exploratory and predictive modeling.

### 2.2 Data Extraction

The FEMA Hazard Mitigation Assistance (HMA) dataset (v4) was obtained from the FEMA Open Data Portal. The original data source is available at (FEMA HMA Mitigated Properties Dataset).

Python tools such as `Pandas` and `NumPy` were used for efficient data manipulation, while `Matplotlib` and `Seaborn` supported visualization. All work was performed in `VS Code` using `Pandas`, ensuring a reproducible and transparent workflow from extraction through modeling.

### 2.3 Data Dictionary

The analysis leverages a subset of fields for focused exploratory data analysis (EDA) and modeling. Table 1 summarizes the selected attributes used in this project.

### 2.4 Data Quality and Cleaning

The dataset is fairly structured but contains known issues like missing values in optional fields, duplicate records, and invalid state entries. For this reason, an additional data cleaning process was carried out, which is explained in detail in Section 3.2 Data Cleaning.

## 3 Methodology

The methodology describes the step-by-step approach followed in this project, covering all stages from data collection to model evaluation.

**Table 1.** Selected fields from the FEMA HMA Mitigated Properties dataset. The table lists the attribute name, description, data type, and an example value for each field.

| **Field Name** | **Description** | **Data Type** | **Example Value** |
|---|---|---|---|
| state | U.S. state where the property is located | String | Texas |
| county | County name where the property is located | String | Harris |
| damageCategory | Severity of damage (e.g., Substantial, Minor) | String | Substantial |
| structureType | Type of property structure (e.g., Single Family) | String | Single Family |
| typeOfResidency | Residency type (e.g., Owner-Occupied) | String | Owner-Occupied |
| foundationType | Property foundation type (e.g., Slab-On-Grade) | String | Slab-On-Grade |
| actualAmountPaid | Amount paid to property owner (USD; excludes administrative costs) | Float | 150000.00 |
| numberOfProperties | Number of properties mitigated in the project | Integer | 5 |
| propertyAction | Mitigation action type (e.g., Elevation, Acquisition) | String | Elevation |
| propertyPartOfProject | Indicates if property is part of a larger project | String | Yes |
| programFy | Fiscal year of the mitigation program | Integer | 2024 |
| date_approved_proxy | Approval date proxy derived from fiscal year | Datetime | 2024-07-01 |
| totalMitigationCost | Total mitigation cost (actualAmountPaid × numberOfProperties) | Float | 750000.00 |

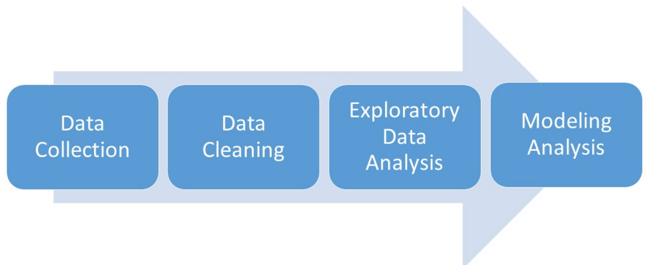


**Fig. 1.** Project implementation workflow,(A. Dhakal, 2025)

## 3.1 Data Collection

The dataset was obtained as a CSV file from the official FEMA Open Data website (FEMA HMA Mitigated Properties Dataset).

## 3.2 Data Cleaning

The data cleaning process transformed the raw FEMA HMA dataset (97,515 rows × 19 columns) into an analysis-ready format, addressing missing values, inconsistencies, duplicates, and irrelevancies. No joins were needed (single-source CSV), but derived features aggregated costs for modeling. Screenshots of pre/post summaries are in Appendix A.

The data cleaning steps included:

1. Early filter to US states and post-2000.
2. Keep relevant columns (including model predictors).
3. Handle missing values early (for categoricals before standardization).
4. Type conversions and standardize categoricals (with str conversion for safety).
5. Derive date proxy and clip negatives.
6. Remove duplicates and derive total cost.
7. Validation and save cleaned data in csv format.

**Missing values:**

Missing data were handled through targeted imputation and selective deletion. For categorical fields such as `damageCategory` (61% nulls), an `Unknown` label preserved 84,026 records. Numerical fields were treated as follows: `numberOfProperties` was imputed with 1, reflecting mostly single-property projects, and rows missing `actualAmountPaid` (48% nulls) were dropped, retaining 47,199 valid cases. These steps align with FEMA data standards and were validated through distribution checks to prevent bias in EDA and modeling.

**Final Dataset**

The final dataset is shown below:

**Table 2.** Summary of Data Cleaning Stages

| Stage | Rows | Columns | Key Action |
|---|---|---|---|
| Raw | 97,515 | 19 | Initial load |
| Filtered | 84,026 | 19 | US states / post-2000 subset |
| Cleaned | 41,308 | 13 | Imputed, derived, and deduplicated |

**Attribute Definitions and Transformations:**
Subset of Table 1, standardized (title-case; mappings applied, e.g., "N/A" → "Unknown"). New fields derived during cleaning:

- `date_approved_proxy`: *datetime* derived from fiscal year.
- `totalMitigationCost`: *float*, calculated as `actualAmountPaid` × `numberOfProperties`.

# 4 Exploratory Data Analysis

The EDA process summarized numeric distributions using histograms, boxplots, and a correlation heatmap, revealing strong right-skewness in *actualAmountPaid*, which was addressed with a log1p transformation. Temporal patterns were examined through fiscal-year aggregations, while mitigation strategies were explored using value counts, pie charts, and bar charts. Geographic concentrations were analyzed through state-level summaries and horizontal bar plots. All visualizations were produced with Seaborn and Matplotlib and saved as PNG files. Overall, the EDA highlighted three core insights: concentrated financial impacts, reactive temporal spending patterns, and FEMA's broader mitigation strategy.

**Log Transformation Code**

```
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

# Apply log1p to handle skewness and zeros
df['log_actualAmountPaid'] = np.log1p(df['actualAmountPaid'])

# Normalized view
sns.histplot(df['log_actualAmountPaid'], kde=True)
plt.title('Log-Transformed Cost Distribution')
```

This transformation reveals underlying patterns hidden by outliers, enabling more reliable visualizations and modeling.

# 5 Modeling and Analysis

The analysis used supervised regression to predict *actualAmountPaid* (log-transformed) based on four categorical features (*state*, *structureType*, *typeOfResidency*, *foundationType*) and one numeric feature (*numberOfProperties*). A `ColumnTransformer` with

`OneHotEncoder` handled preprocessing, and the data was split 80/20 for training and testing. Two models were evaluated: a baseline Linear Regression and a Random Forest Regressor designed to capture non-linear interactions in the one-hot encoded feature space. Performance was assessed using $R^2$ on the log scale and MAE/RMSE on the original scale. The model with the higher log-scale $R^2$ was selected for state-level projections to estimate FY2030 outcomes, supported by bar and line visualizations.

**Code Example: Model Training and Evaluation**

Random Forest Model Training

```
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import r2_score

rf_model = RandomForestRegressor(
    n_estimators=100,
    max_depth=15,
    random_state=42
)

rf_model.fit(X_train, y_train)  # y_train = log_costs
y_pred_log = rf_model.predict(X_test)

r2_log = r2_score(y_test, y_pred_log)
# High R² indicates strong fit
print(f"Random Forest R² (Log): {r2_log:.4f}")
```

Random Forest's ensemble approach captures non-linear cost drivers, outperforming linear baselines for accurate projections.

# 6 Results

All results presented here are based on the Python notebooks for this project, with all code available at: [GitHub Notebook Link].

Disasters are increasing, and FEMA's HMA program provides protection, but spending is uneven-focused on high-risk areas, primarily homes, and surging after major events. The analysis below examines how costs, strategies, timing, and geography shape national resilience.

## 6.1 Uncovering Extreme Cost Concentration and Model Preparation

The distribution of money paid by FEMA reveals the impacts of major disasters and their associated costs driving the costs higher by those disasters. The average project cost ($272, 325) is four times the 75th percentile ($66, 046) (Table 3). This extreme concentration validates the focus of the project: a small fraction of high-cost projects drives the entire mitigation budget. Although majority of the projects cost 0 to 100 million dollars, there are a few which are significantly expensive with the cost almost a billion dollars (Fig.2). Even the cheaper projects are not cheap as the graph in fig.2

shows the values in hundred millions ($10^8$). Most of the projects cost one hundred to two hundred million dollars.

**Table 3.** Initial descriptive statistics for the pre-modeling dataset. Shows count, mean, and quartiles for key variables including Actual Amount Paid, Number of Properties, and Total Mitigation Cost.

| Variable | Count | Mean | 25% | 75% |
|---|---|---|---|---|
| Actual Amount Paid ($) | 41,308 | 272,325 | 3,972 | 66,046 |
| Number of Properties | 41,308 | 1.58 | 1 | 1 |
| Total Mitigation Cost ($) | 41,308 | 299,917 | 4,703 | 69,500 |

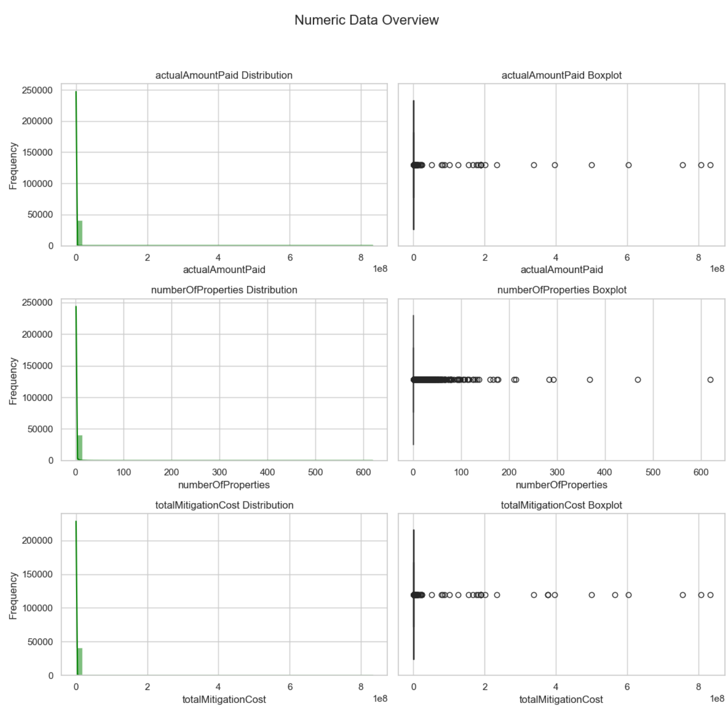


**Fig. 2.** Distribution of Actual Amount Paid, Number of Properties, and Total Mitigation Cost.Histograms and boxplots show strong right-skewness across all three variables, with most projects concentrated at lower cost ranges and a long tail extending into multi-million-dollar amounts. The presence of substantial outliers further supports applying a target variable transformation.(A. Dhakal, 2025)

**Correlation Analysis:**

A correlation heatmap was generated to inspect the relationships between the numeric features before modeling. Fig 3. Correlation Heatmap of Numeric Variables suggests that the actual amount paid is strongly correlated to totalmitigation cost but it is not strongly correlated to number of properties. It could be because of the fact that there are fewer numbers of properties that cost a lot and majority of properties do not cost high amounts in different states.

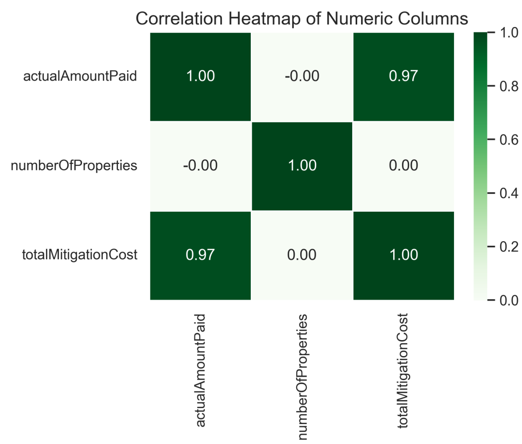


**Fig. 3.** Correlation heatmap of numeric variables, showing strong positive correlation between Actual Amount Paid and Total Mitigation Cost and low correlation with Number of Properties.(A. Dhakal, 2025)

**Target Transformation:**

The target variable exhibited extreme right-skewness, a normalization step was required for robust regression modeling. A $\log(x+1)$ transformation was applied to stabilize the variance and reduce the influence of outliers.

**Code Example (Transformation):**

Log Transformation

```
import numpy as np
# Applying log(x + 1) transformation
df['log_cost'] = np.log1p(df['actualAmountPaid'])
```

The transformation successfully converted the highly skewed target into an approximately normal distribution, enabling more stable and reliable model training.

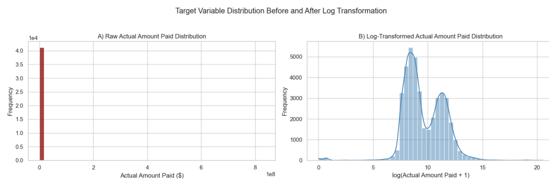


**Fig. 4.** Target variable distribution before and after log(x + 1) transformation. Log transformation reduces skewness and stabilizes variance for regression modeling.(A. Dhakal, 2025)

## 6.2 Mitigation Strategy: Dual Approaches to Risk

Frequency analysis was used to determine FEMA's strategic priorities by propertyAction and structureType. Most of the mitigation funding is concentrated on certain types of risk-reduction strategies. Wind Retrofit and Acquisition/Demolition account for more than 70% of all spending. Most of the mitigation budgets are spent in supporting Single Family structures.

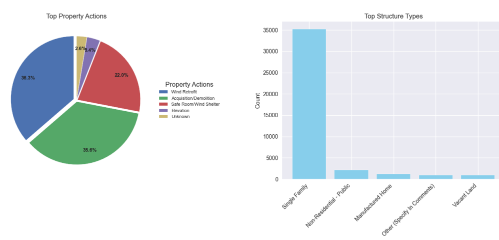


**Fig. 5.** Distribution of mitigation actions by propertyAction and structureType. Wind retrofits and acquisition/demolition dominate, with single-family homes receiving most funding.(A. Dhakal, 2025)

## 6.3 Funding Trends: Peaks After Disaster

Federal funding by FEMA spikes in response to major disaster. From Fig.6, the fiscal years 2008, 2011, and 2017 show a high increase in both total dollars paid and properties mitigated. These years correspond to major disasters in recent U.S. history, including post-Hurricane Katrina recovery and Midwest floods (2008), the Joplin tornado and Mississippi River floods (2011), and Hurricanes Harvey, Irma, and Maria (2017).

**Code Example:**

Yearly Aggregation

```
yearly_trends = df.groupby('programFy').agg({
    'numberOfProperties': 'sum',
    'actualAmountPaid': 'sum'
}).reset_index()
```

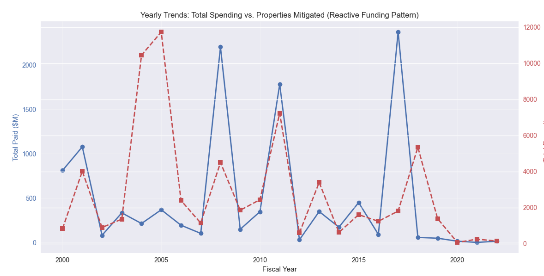


**Fig. 6.** Yearly funding trends showing reactive spikes after major disasters. Peaks follow post-disaster years, indicating FEMA's mitigation is largely event-driven.(A. Dhakal, 2025)

## 6.4 Geographic Concentration: High-Cost Hotspots

All of the states receive one form or another mitigation dollars based on the disasters they encounter. Among them Missouri ($2.6B), Washington ($1.7B), and Florida ($1.1B) account for 45 percent of all dollars spent (Fig. 7). This result suggests targeted investment based on the repeated disasters they encounter. These investments also raise questions about balancing urgent need in some particular areas in response to a disaster versus developing capacity for the entire nation.

These hotspots reflect regional hazards:

- Missouri's large-scale flood buyouts
- Washington's seismic retrofits
- Florida's hurricane-hardening efforts

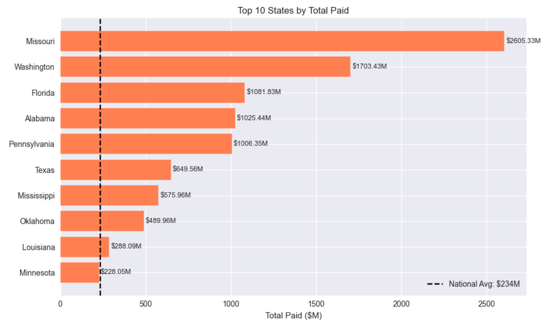


**Fig. 7.** Top 10 states by total mitigation spending. Missouri, Washington, and Florida receive the largest share, reflecting regional hazard exposure.(A. Dhakal, 2025)

## 6.5 Modeling and Analysis

### 6.5.1 Predictive Model Setup and Comparison

To address the wide range in project costs, the target variable was transformed using $\log(\text{actualAmountPaid} + 1)$. This transformed outcome was modeled using key features including state, structureType, foundationType, numberOfProperties, and `programFy`. Categorical variables were one-hot encoded, and numerical variables were used as-is. Two predictive models-Linear Regression and Random Forest Regressor-were trained and compared using the $R^2$ score on the log-transformed test data. An 80/20 train–test split (random state = 42) ensured consistent evaluation. Random Forest achieved higher $R^2$ values for both training and test sets (Table 4). R² values near zero on the original scale do not indicate poor performance. They simply reflect the extreme right-skew of the cost data. The log transformation is necessary to stabilize variance, reduce outlier influence, and provide a more meaningful basis for evaluation.

**Feedback Note**

Albert Kabore: " Anjana has completed an excellent capstone with clear writing, strong visuals, and a thoughtful modeling approach. If she chooses to enhance anything, she might add a short explanation in the Modeling section clarifying why the R² on the original scale appears near zero even though the log scale R² is strong. She already reports both scales correctly in her notebook and paper, so a brief interpretive note would help readers understand that log-scale performance is the appropriate evaluation metric for extremely skewed cost data. This small addition would make her already strong modeling section even more accessible to a broad audience".

**Table 4.** Model performance comparison for LR and RFR. Metrics include R², MAE, and RMSE on training and test datasets (log-transformed and original).

| Dataset | Model | $R^2$ (Log) | $R^2$ (Original) | MAE ($) | RMSE ($) |
|---|---|---|---|---|---|
| Training | Linear Regression | 0.3955 | -0.0006 | 265,200 | 9,092,721 |
| Training | Random Forest Regressor | 0.6570 | 0.0019 | 253,640 | 9,081,705 |
| Test | Linear Regression | 0.3857 | -0.0003 | 222,836 | 9,635,366 |
| Test | Random Forest Regressor | 0.5847 | -0.0003 | 213,165 | 9,634,923 |

**Table 5.** Five-fold cross-validation summary for LR and RFR. Shows mean and standard deviation of R² on log-transformed data and test R².

| Model | Mean CV $R^2$ (Log) | Std CV $R^2$ (Log) | Test $R^2$ (Log) |
|---|---|---|---|
| Linear Regression | 0.3858 | 0.0112 | 0.3857 |
| Random Forest Regressor | 0.5032 | 0.0105 | 0.5847 |

### 6.5.2 Factors Impacting the Mitigation Costs

Program Fiscal Year was the strongest predictor (0.13), indicating mitigation costs trend upward over time. The State of Oklahoma followed (0.12), likely reflecting tornado-related disasters. Foundation types and the number of properties contributed less.

### 6.5.3 Top 5 State Projection (RFR Model)

The Random Forest Regressor predicted mitigation costs for the full dataset and for a projected future scenario where *programFy* was increased by five years (simulating FY2030). Predictions were inverse-transformed to USD and aggregated by state. National costs rose 17.9% (\$1.63B $\rightarrow$ \$1.92B). However, linear trends for the top 5 states showed sharp declines (Table 6, Fig. 8), possibly reflecting lower recent spending compared with major historical events.

**Table 6.** Top 5 states by historical and projected FY2030 mitigation spending (RFR). Shows declines in historically high-spending states, indicating saturation effects.

| State | Current Cost ($M) | Projected FY2030 Cost ($M) |
|---|---|---|
| Missouri | 2,605 | 240 |
| Washington | 1,703 | 19 |
| Florida | 1,082 | 251 |
| Alabama | 1,025 | 107 |
| Pennsylvania | 1,006 | 62 |

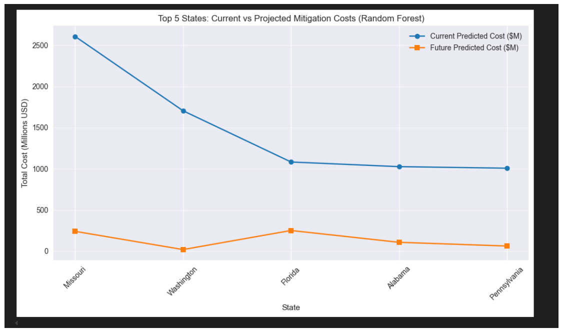


**Fig. 8.** Current versus projected FY2030 mitigation spending for the top 5 states. Declines in historically high-spending states suggest saturation effects and potential funding shifts.(A. Dhakal, 2025)

## 7 Conclusions and Discussion

This analysis of more than 41,000 FEMA Hazard Mitigation Assistance (HMA) projects from 2000-2025 shows that almost 1.63 billion USD is spent in responding to hazards and on mitigation costs in the U.S. A majority of such projects are small but there are a few huge projects which skew the results. Majority of such budgets seem to be spent on a few categories like wind retrofits and property acquisitions/demolitions. These two categories account 70% of all activities (Fig. 5). One hightlights of thee result is a focus on single-family homes. This suggests that the projects help working family to be more resilient and recover from the disasters.

Funding trends highlight the reactive nature of the FEMA's assistance. The large spending correspond to the major disaster periods like 2008, 2011, and 2017. These periods relate to post-disaster periods following Hurricanes Katrina, Sandy, and Harvey/Irma (Fig. 6). The geographic pattern adds another imbalance: Missouri ($2.6B), Washington ($1.7B), and Florida ($1.1B) account for 45% of total spending (Fig. 7), leaving many regions comparatively underserved. These disparities raise equity concerns as disaster risks grow nationwide.

The Random Forest Regressor (RFR) provides further insight by capturing nonlinear relationships that linear regression misses (test R2 = 0.5847 vs. 0.3857; Table 5). The model identifies fiscal year (0.1285) and state-specific factors (e.g., Oklahoma at 0.1222) as the strongest cost drivers, while project count plays a smaller role (0.0613). Cross-validation supports model stability (mean R2 = 0.5032; Table 6). Forecasts suggest a 17.9% increase in national spending by FY2030 (from $1.63B to $1.92B), paired with sharp slowdowns in historically high-investment states: projected declines of 91% in Missouri and 99% in Washington (Table 7; Fig. 8). This pattern reflects saturation in high-spend areas and steady national growth, emphasizing FEMA's challenge of balancing episodic surges with sustained, equitable investment.

In summary, HMA effectively addresses immediate crises but falls short on proactive prevention, leaving communities vulnerable. These findings call for a shift toward data-driven, equitable strategies that turn episodic relief into lasting resilience [4].

### 7.1 Limitations

Here are some of the limitations of this study:

1. It includes data reported, maintained, and made publicly available by FEMA only.
2. The FEMA dataset is heavily weighted toward residential projects (over 70%), which may underrepresent public or commercial infrastructure needs.
3. Data cleaning removed 57% of original records (from 97,515 to 41,308), largely due to extensive missing fields (e.g., 59,807 null values in `damageCategory`), potentially excluding lower-severity projects.
4. The RFR model also assumes temporal stability, which may not hold true as the disasters are rarely linear in occurence.

### 7.2 Recommendations

1. **Shift to proactive funding:** Increase HMA support during non-disaster years to help low-funding states and reduce risks before crises occur.
2. **Diversify project focus:** Continue supporting residential work but expand investment in public and commercial infrastructure, which remains underfunded.
3. **Risk-based allocation:** Apply Random Forest insights-especially the strong influence of state and foundation type to design risk-indexed grants that account for future exposure and project complexity rather than relying solely on past losses.

## 8 References

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4. Rose, A., et al. (2007). *Benefit-Cost Analysis of FEMA Hazard Mitigation Grants*. *Natural Hazards Review*, 8(4), 97–111.

## 9 Appendix

## Supplementary Materials

The analysis, cleaned datasets, and code are available at the following locations:

1. Overleaf link :Overleaf Project
2. GitHub Repo link: GitHub Repo

## Appendix A: Pre- and Post-Cleaning Dataset Summaries

This appendix provides textual representations of key pre- and post-cleaning summaries from the Jupyter Notebook outputs. These include dataset info, null value counts, and data types, generated via Pandas `.info()`, `.isnull().sum()`, and `.dtypes`.

## A.1 Raw Dataset Summary (Pre-Cleaning)

**Dataset Shape:** 97,515 rows × 19 columns

**Data Overview:** The dataset contains 97,515 entries, indexed from 0 to 97,514.

**Column Information and Data Types:**

| # | Column | Non-Null Count | Dtype |
|---|---|---|---|
| 0 | projectIdentifier | 97515 (non-null) | obj |
| 1 | programArea | 97515 (non-null) | obj |
| 2 | programFy | 97515 (non-null) | int |
| 3 | disasterNumber | 85184 (non-null) | flt |
| 4 | propertyPartOfProject | 93361 (non-null) | flt |
| 5 | propertyAction | 90837 (non-null) | obj |
| 6 | structureType | 90427 (non-null) | obj |
| 7 | typeOfResidency | 61502 (non-null) | obj |
| 8 | foundationType | 87029 (non-null) | obj |
| 9 | county | 94453 (non-null) | obj |
| 10 | city | 97511 (non-null) | obj |
| 11 | state | 97477 (non-null) | obj |
| 12 | stateNumberCode | 97476 (non-null) | flt |
| 13 | region | 97477 (non-null) | flt |
| 14 | zip | 97505 (non-null) | obj |
| 15 | damageCategory | 37708 (non-null) | obj |
| 16 | actualAmountPaid | 50352 (non-null) | flt |
| 17 | numberOfProperties | 97515 (non-null) | int |
| 18 | id | 97515 (non-null) | obj |

**Null Value Counts (Sorted Descending):**

| Column | Null Count |
|---|---|
| damageCategory | 59,807 |
| actualAmountPaid | 47,163 |
| typeOfResidency | 36,013 |
| disasterNumber | 12,331 |
| foundationType | 10,486 |
| structureType | 7,088 |
| propertyAction | 6,678 |
| propertyPartOfProject | 4,154 |
| county | 3,062 |
| stateNumberCode | 39 |
| region | 38 |
| state | 38 |
| zip | 10 |
| city | 4 |
| projectIdentifier | 0 |
| programFy | 0 |
| programArea | 0 |
| numberOfProperties | 0 |

| | |
|---|---|
| id | 0 |

## A.2 Cleaned Dataset Summary (Post-Cleaning)

**Final Dataset Shape:** 41,308 rows × 13 columns

**Column Information and Data Types:**

| Column | Dtype |
|---|---|
| state | obj |
| county | obj |
| damageCategory | obj |
| structureType | obj |
| typeOfResidency | obj |
| foundationType | obj |
| actualAmountPaid | flt |
| numberOfProperties | int |
| propertyAction | obj |
| propertyPartOfProject | obj |
| programFy | int |
| date_approved_proxy | dt |
| totalMitigationCost | flt |

**Null Value Counts (All Zero):**

| Column | Null Count |
|---|---|
| state | 0 |
| county | 0 |
| damageCategory | 0 |
| structureType | 0 |
| typeOfResidency | 0 |
| foundationType | 0 |
| actualAmountPaid | 0 |
| numberOfProperties | 0 |
| propertyAction | 0 |
| propertyPartOfProject | 0 |
| programFy | 0 |
| date_approved_proxy | 0 |
| totalMitigationCost | 0 |